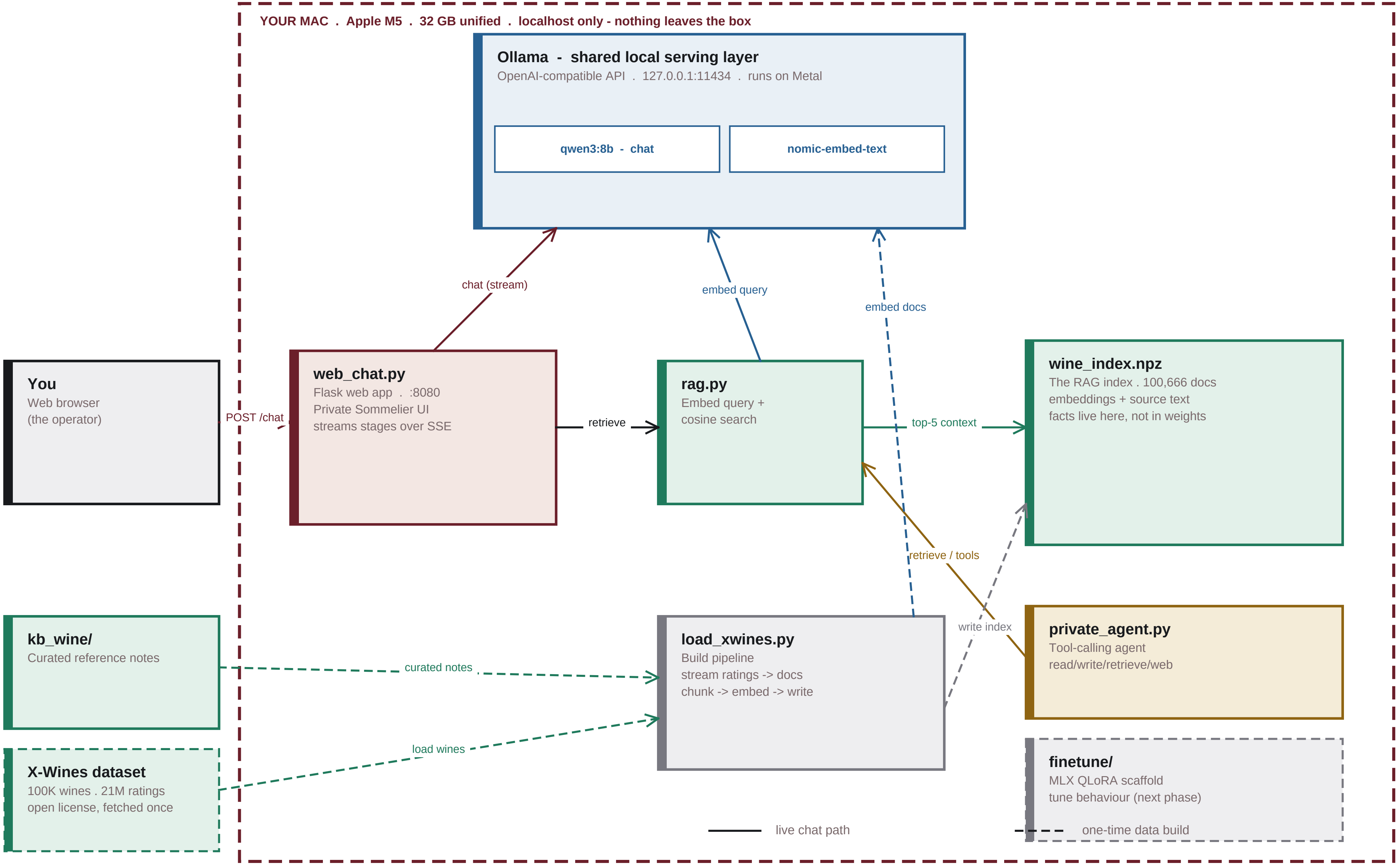
# Private LLM - System Context Diagram

A private wine assistant on Apple Silicon. Every box runs locally and binds to 127.0.0.1; Ollama is a shared local service. The X-Wines dataset and model weights are fetched once, then it all runs offline.



# What This System Does

A plain-language tour of the private wine assistant and why it is built the way it is.

## Overview

A private wine sommelier that runs entirely on your Mac. You ask questions in a browser; a local language model answers, grounded in a wine knowledge base of 100,666 entries. No cloud, no API key, and no data leaves the machine.

## Sovereignty by construction

Every component binds to 127.0.0.1. The model runs on Apple Silicon (Metal) through Ollama; retrieval and the web app are local processes. The only outbound step is a one-time download of the open X-Wines dataset and the model weights - after that it runs fully offline.

## Two data layers

Curated knowledge (kb_wine/) explains how wine works: regions, grapes, classifications, pairing. The X-Wines dataset adds 100K real bottles with average ratings drawn from 21M reviews. Both are embedded into one index, so a single question can pull on reference knowledge and specific recommendations at once.

## Facts in retrieval, not weights

The model supplies language and reasoning; the facts come from the index at query time. Update a fact by editing a file and re-indexing - no retraining, and every answer stays auditable against its sources.

## The components

- Ollama - serves qwen3:8b (chat) and nomic-embed-text (embeddings) on :11434.
- rag.py - embeds the question and does cosine search over the index.
- wine_index.npz - the embedded knowledge base (100,666 documents).
- web_chat.py - Flask web UI on :8080; streams its progress over SSE.
- load_xwines.py - builds the index from kb_wine/ and the X-Wines dataset.
- private_agent.py - a sandboxed tool-calling agent (read/write/retrieve/web).
- finetune/ - an MLX QLoRA scaffold to tune behaviour (the next phase).

## A question, end to end

Your question is embedded, matched against the index, and the top wines become context for the model, which streams an answer back to the browser with its sources. You watch each stage live: searching the cellar, the wines it found, then the answer being written.

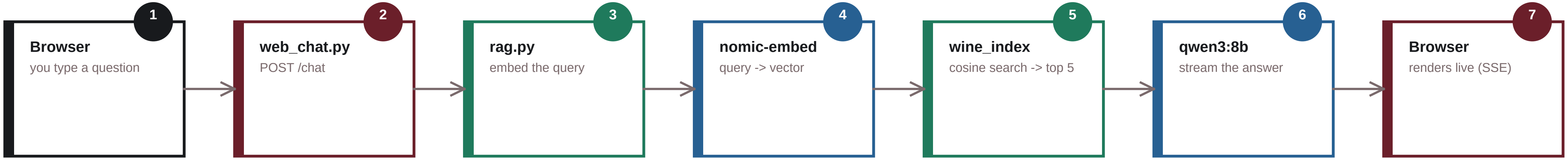
## What's next

Swap in a larger local model (qwen3:14b or 30b-a3b), run the fine-tune, or load your own cellar and tasting notes into kb_wine/ to personalise the recommendations.

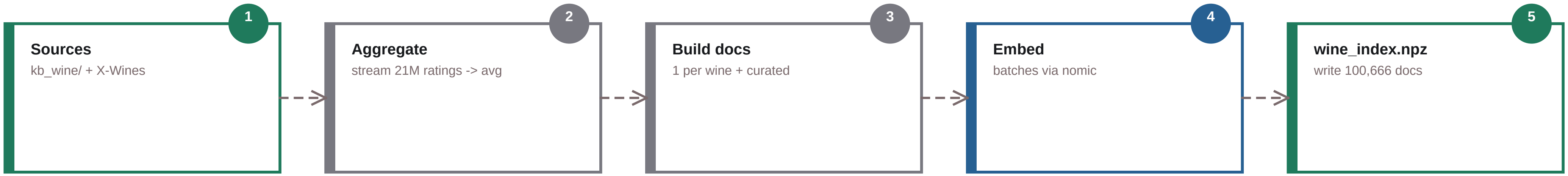
# Ins & Outs - How Data Flows

Two paths move through the system: answering a question (live) and building the knowledge (one-time).

## Flow A - Answering a question (live, every request)




## Flow B - Building the index (one-time, re-run when data changes)




**INPUTS (what goes in)**
Your questions (typed in the browser).
kb_wine/ - curated reference notes you write.
X-Wines dataset - 100K wines / 21M ratings, fetched once.
Model weights - qwen3:8b + nomic-embed-text, pulled once.

**OUTPUTS (what comes out)**
Streamed answers with their sources (in the browser).
wine_index.npz - the reusable embedded knowledge base.
Agent: files written into its sandbox (when enabled).
Fine-tune: LoRA adapters under finetune/ (next phase).